

ASSIGNMENT-4

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CSE-D

- ① Discuss the ANN - genetic hybrid computing technique in detail. Also discuss the implementation with an example.

(A) ANN - genetic hybrid computing technique

This combination of the strengths of artificial neural networks (ANN) and genetic algorithms (GA) is an intelligent way of approaching complex optimization and learning problems.

Besides this hybridization has the advantage of utilizing the learning ability of ANN along with the optimization strengths of GA.

Artificial Neural Network (ANN)

Artificial neural networks are a computing model based on the structure and functioning of an artificial brain. They can learn the patterns and predict outcomes.

They are useful for many classification and regression problems and pattern recognition.

Tuning ~~key~~ hyperparameters such as network architecture and learning rates can take much trial and error and

sometimes the models can get into local minima during training

Genetic Algorithm (GA)

GAs are search heuristics based on the principles of natural selection and evolution.

They are excellent for optimizing complex functions and exploring large solution spaces.

They evolve a multilevel population of candidate solutions through selection, crossover and mutation toward an optimal or near optimal solution.

Hybrid Approach

By merging them together, the respective shortcomings of ANN and GA counterbalance each other.

The hybrid system is more robust than either can reach stuck into local minima quickly and can stimulate dynamic environments.

Implementation Steps

(1) Initialization

→ represent the parameters of the ANN (eg- weights or biases) as chromosomes.

→ randomly initialize a population of these chromosomes.

(II) Fitness Evaluation

- each chromosome is decoded to construct a neural network.
- Train the ANN using a subset of training data and compute its fitness based on a defined metric

(III) Selection

Select the best performing chromosomes for reproduction using techniques like roulette wheel or tournament selection.

(IV) Crossover

Combine pairs of parent chromosomes to produce offspring, inheriting characteristics from both parents.

(V) Mutation

Apply random changes to offspring chromosomes to introduce genetic diversity.

(VI) Replacement

Replace the worst offspring individuals in the population with new offspring.

(VII) Final ANN

Decode the best chromosome to construct the optimized ANN and evaluate its performance on the test data.

Example - ANN genetic hybrid for handwritten digit recognition (MNIST)

(i) Problem

Train a neural network to classify handwritten digits from the MNIST dataset.

(ii) Implementation

Chromosome encoding - each chromosome represents ANN weights and biases as a flattened array

Fitness function - accuracy on a validation set.

GA steps -

- Initialize a population of random weights
- evaluate fitness using ANN accuracy.
- perform selection, crossover and mutation to create new weights.

Train the ANN for a few epochs on the training set and re-evaluate fitness after updates.

(iii) Performance

This hybrid approach is particularly useful when backpropagation struggles, for instance with vanishing gradients or poor initialization.

(2) (a) Define and differentiate between the hybrid techniques.

(A) Hybrid techniques in computing refer to methods that combine two or more computational paradigms, algorithms or models to achieve better performance, adaptability or efficiency than individual methods could offer.

For example,

* ANN-genetic algorithm hybrid - combines artificial neural networks for learning and genetic algorithms for optimization.

* Fuzzy neural hybrid - Integrates fuzzy logic for handling uncertainty with neural networks for learning and adaptability.

Aspect	Sequential hybrid	Integrated hybrid	Parallel hybrid
Execution flow	Methods are applied in a sequential manner.	Methods work together in a unified manner	Methods run concurrently & share results
Interaction level	Minimal.	High	Moderate
Complexity	Low	High	Moderate
Examples	GA for ANN initialization	Fuzzy logic within ANN.	ANN & reinforcement learning for control

(b) Briefly describe how neural networks and genetic algorithms operate independently. What are the key strengths and limitations of each?

(A) How neural networks operate

(I) Structure - Consists of layers of interconnected nodes
→ input layer for data.
→ hidden layer for feature extraction and transformation.

(II) Training - Neural networks use backpropagation and gradient descent to adjust weights and biases to minimize a loss function.

(III) Learning - Neural networks learn from labelled or unlabelled data through iterative weight updates.

~~III~~

How genetic algorithm operates

(I) Initialization - Start with a population of candidate solutions (chromosomes), each represented as a string.

(II) Selection - Choose the fittest individuals based on a fitness function.

(III) Crossover - Combine genetic material from two parents to create offspring with shared traits.

(iv) Mutation - Introduce random changes in chromosomes to maintain diversity and avoid premature convergence.

(v) Iteration - Repeat selection, crossover and mutation over generations until an optimal or near optimal solution is found.

Aspect	Neural Networks	Genetic Algorithms
Primary focus	learning patterns from data	optimizing solutions in search spaces.
Training approach	Gradient based optimization	Evolutionary processes
Data dependency	Requires substantial data for training	Works without direct data on objective function
Strength	Good for generalizing from examples	Effective for global optimization
Limitation	Sensitive to initialization and tuning	Computationally expensive for large problems.